FINAL YEAR PROJECT · BSC (HONS) SOFTWARE DEVELOPMENT · SETU

A Machine Learning Approach to a Real-Time Network Intrusion Detection System

A Random Forest + XGBoost ensemble that classifies live network flows into 15 attack categories, with per-prediction SHAP explanations, Prometheus monitoring, and one-command Docker deployment.

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ML-Based-NIDS

99.4%	91.9%	15	2.1M
TEST ACCURACY	BALANCED ACC.	ATTACK CLASSES	FLOWS TRAINED

01 · THE PROBLEM

Signature-based IDS cannot catch what it has not already seen.

Traditional intrusion detection systems rely on hand-written rules that match known attack signatures — meaning zero-day and polymorphic attacks slip through. ML-based detectors can generalise, but the field has a trust problem: most models are opaque "black boxes" that security teams cannot audit or justify to an auditor. This project closes that gap.

SOLUTION ARCHITECTURE

Scapy Live Capture
BPF filter · thread-safe queue

↓

Flow Aggregator
5-tuple bidirectional flows

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Feature Extractor
84 CIC-IDS2017 statistics

↓

FastAPI · /predict
scaler · SelectKBest (84→50)

↓

RF + XGBoost Soft-Voting
+ SHAP TreeExplainer top-5

↓

Alert · Grafana · SSE Dashboard

WHY THIS DESIGN

Decoupled tiers — stateless classifier API is language-agnostic and independently benchmarkable.

Tree ensemble — tabular flow features don't need deep learning; trees train faster and admit exact SHAP.

Threaded predictions — ThreadPoolExecutor keeps capture running during 10–100 ms API round-trips.

SMOTE + class weights — addresses severe class imbalance in CIC-IDS2017.

HEADLINE RESULTS

MODEL	ACCURACY	MACRO-F1
XGBoost	**99.44%**	0.834
Random Forest	**99.40%**	0.873
Ensemble (voting)	**99.4%**	0.86

13 of 15 attack classes achieve ≥80% class accuracy on a held-out test set of 311,746 flows; randomised evaluation 91.9% ± 1.2% (95% CI).

02 · TECHNOLOGY STACK

Python 3.11 · scikit-learn · XGBoost · SHAP · Scapy · FastAPI · Flask (SSE) · Prometheus · Grafana · Docker Compose · CIC-IDS2017 · SMOTE

03 · KEY CONTRIBUTIONS

WHAT'S NOVEL

End-to-end pipeline from raw packets to explainable verdict — not just a model on static CSVs.

Per-prediction SHAP top-5 attributions make every alert auditable by a human analyst.

EVALUATION RIGOUR

Randomised evaluation with 95% CI bounds, not a single lucky split.

Live-traffic demo replays CIC-IDS2017 PCAPs against the running API to prove wire-to-alert latency.

Production-shaped — rate-limited API, metrics, Grafana dashboard, reproducible Docker stack.

Confusion matrix & SHAP global importance reported per class — failure modes made visible.

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REPO ML-Based-NIDS

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→ Scan the QR code for the full source, documentation, and report.

View Source →